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Rothschild Experimentation in Two-Armed Bandits with Costly Risky Learning

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Introduction and background on bandits and exploration-exploitation

Bandit models study a basic choice problem under uncertainty: whether to use the option that looks best now or try another option to learn more. The two-armed bandit is the simplest version and is the main starting point for work on experimentation, risky learning, and costly information acquisition. (8 sources)

A useful starting point is the basic exploration-exploitation problem. In a bandit setting, a decision maker repeatedly chooses among actions with uncertain payoffs, and each choice affects both current reward and what can be learned for the future. The core tension is between taking the arm that is already better understood and trying a less-known arm that may pay more later. This is the standard way the literature frames experimentation. (Peluso et al., 2017) (Yoshida et al., 2015) (Yu, 2011)

The two-armed bandit is the canonical minimal case. One can think of it as choosing at each date between two stochastic processes or two arms, observing a reward, and then updating beliefs about uncertain payoffs over time. In Bayesian versions, beliefs about the risky or unknown arm are revised after each observation, so action choice and learning are tightly linked. The objective is usually to maximize expected cumulative reward over a finite or indefinite horizon. (Yu, 2011) (Zhang et al., 2025)

This framework matters because myopic payoff maximization is often not optimal. A choice that gives a lower payoff now can still be best if it generates useful information that improves later decisions. That is exactly why bandit models became the standard tool for studying experimentation in economics and related fields. (Peluso et al., 2017) (Yu, 2011)

The same logic also extends beyond a fixed and static environment. In restless bandits, payoffs can change over time, which makes learning harder because old information can become stale. Work on social versus asocial learning uses this setting to show that information can come either from one's own experimentation or from observing others, with social learning often cheaper but less reliable because it may be delayed or outdated. (Yoshida et al., 2015) (Roche et al., 2010) (Toyokawa et al., 2014)

Finally, even simple-looking bandit problems can become hard once one allows many arms, changing states, or richer horizons. Experimental work still often returns to two- or three-armed settings precisely because they isolate the basic learning problem in a clean way, while theory has long stressed that richer restless cases can be computationally difficult. (Hudja et al., 2025) (Yoshida et al., 2015) (Papadimitriou et al., 1994)

For the user's query, this background sets up the Rothschild-style experimentation problem: in the simplest two-armed case, one arm is often safe and the other risky but informative, so the value of trying the risky arm comes not just from its current payoff but from the information it produces about future choices. (Model-Generated)

Evidence

(Peluso et al., 2017)

Learning vs earning trade-off with missing or censored observations: The two-armed Bayesian nonparametric beta-Stacy bandit problem

> *"This selection is made on the basis of the previous observations, and it balances two conflicting benefits: the immediate payoff coming from the exploitation of a better known arm and the information concerning future payoffs coming from the exploration of a less known arm."*

(Yoshida et al., 2015)

Interactive Restless Multi-armed Bandit Game and Swarm Intelligence Effect

> *"The trade-off between the exploitation of good choices and the exploration of unknown but potentially more profitable choices is a well-known problem (Kameda et al., 2003)(Roche et al., 2010)(Gueudré et al., 2013) . A multi-armed bandit (MAB) provides the most typical environment for studying this trade-off. It is defined by sequential decision making among multiple choices that are associated with a payoff. The MAB problem involves the maximization of the total reward for a given period or budget"*

> *".Recently, the MAB has also provided a good environment for the tradeoff between social and asocial learning (Roche et al., 2010) . Here, social learning is learning through observation or interaction with other individuals, and asocial learning is individual learning (Jones, 1987)(Roche et al., 2010)(Kameda et al., 2003)(Toyokawa et al., 2014) . The advantage of social learning is its cost compared with asocial learning. The disadvantage is its error-prone nature, as the information obtained by social learning might be outdated or inappropriate"*

> *".Here, restless means that the payoff of each bandit changes over time. There are 100 bandits in an rMAB, and each bandit has a distinct payoff independently drawn from an exponential distribution. The probability that a payoff changes per round is p c . An agent has three options for each round: Innovate, Observe, and Exploit. Innovate and Observe correspond to asocial and social learning, respectively. For Innovate, an agent obtains the payoff information of one randomly chosen bandit. For Observe, an agent obtains the payoff information of n O randomly chosen bandits that were exploited by the agents during the previous round. Compared to the information obtained by Innovate, that obtained by Observe is older by one round. For Exploit, an agent chooses a bandit that he has already explored by Innovate or Observe and obtains a payoff. In an rMAB environment, it is extremely difficult for agents to optimize their choices (Papadimitriou et al., 1994)(Roche et al., 2010) ."*

(Yu, 2011)

Structural Properties of Bayesian Bandits with Exponential Family Distributions

> *"At each of n stages, an experimenter must take an observation from one of two stochastic processes (arms). Let us adopt the Bayesian framework and assume that the experimenter's belief about an unknown arm is updated according to Bayes Theorem after each observation"*

> *".This is a finite-horizon two-armed bandit (Berry and Fristedt 1985), a classical problem in sequential decision theory"*

> *".Bandit problems exhibit a well-known exploration-exploitation tradeoff. Simply maximizing the immediate payoff is usually not an optimal strategy; one must allow for exploring an unknown arm for higher payoff later on."*

(Zhang et al., 2025)

Strategic A/B testing via Maximum Probability-driven Two-armed Bandit

> *"The two-armed bandit is a classic and widely studied model in reinforcement learning, which consists of three elements: the agent, the action space, and the reward space. Let ϑ i ∈ {0, 1} represent the two arms of the bandit, where i denotes the time step. At each step, the agent selects an arm ϑ i and obtains an independent reward R (ϑi) i"*

> *"the goal of the two-armed bandit is to find an optimal policy θ * n = (ϑ * 1 , . . . , ϑ * n) that maximizes the expected cumulative reward."*

(Roche et al., 2010)

Why Copy Others? Insights from the Social Learning Strategies Tournament

"It Pays to Be a Copy Cat Does it pay to copy what others do? Rendell et al. (p. 208) elected to copy Robert Axelrod's 1979 tournament in which strategies for playing the iterated prisoner's dilemma game were pitted against each other until an overall winner emerged—the tit-for-tat strategy. In the 2008 tournament, 100 social learning strategies designed to cope with a changing environment competed against each other; the winning strategy involved sampling the behaviors of other players periodically, rather than exploring the environment alone. Learning from what others do is more efficient than learning all on one's own. Social learning (learning through observation or interaction with other individuals) is widespread in nature and is central to the remarkable success of humanity, yet it remains unclear why copying is profitable and how to copy most effectively. To address these questions, we organized a computer tournament in which entrants submitted strategies specifying how to use social learning and its asocial alternative (for example, trial-and-error learning) to acquire adaptive behavior in a complex environment. Most current theory predicts the emergence of mixed strategies that rely on some combination of the two types of learning. In the tournament, however, strategies that relied heavily on social learning were found to be remarkably successful, even when asocial information was no more costly than social information. Social learning proved advantageous because individuals frequently demonstrated the highest-payoff behavior in their repertoire, inadvertently filtering information for copiers. The winning strategy (discountmachine) relied nearly exclusively on social learning and weighted information according to the time since acquisition."

(Toyokawa et al., 2014)

Human Collective Intelligence under Dual Exploration-Exploitation Dilemmas

"The exploration-exploitation dilemma is a recurrent adaptive problem for humans as well as non-human animals. Given a fixed time/energy budget, every individual faces a fundamental trade-off between exploring for better resources and exploiting known resources to optimize overall performance under uncertainty. Colonies of eusocial insects are known to solve this dilemma successfully via evolved coordination mechanisms that function at the collective level. For humans and other non-eusocial species, however, this dilemma operates within individuals as well as between individuals, because group members may be motivated to take excessive advantage of others' exploratory findings through social learning. Thus, even though social learning can reduce collective exploration costs, the emergence of disproportionate "information scroungers" may severely undermine its potential benefits. We investigated experimentally whether social learning opportunities might improve the performance of human participants working on a "multi-armed bandit" problem in groups, where they could learn about each other's past choice behaviors. Results showed that, even though information scroungers emerged frequently in groups, social learning opportunities reduced total group exploration time while increasing harvesting from better options, and consequentially improved collective performance. Surprisingly, enriching social information by allowing participants to observe others' evaluations of chosen options (e.g., Amazon's 5-star rating system) in addition to choice-frequency information had a detrimental impact on performance compared to the simpler situation with only the choice-frequency information. These results indicate that humans groups can handle the fundamental "dual exploration-exploitation dilemmas" successfully, and that social learning about simple choice-frequencies can help produce collective intelligence."

(Hudja et al., 2025)

Strategies in the multi-armed bandit

"This paper analyzes individual behavior in multi-armed bandit problems"

". We use a between-subjects experiment to implement four bandit problems that vary based on the horizon (indefinite or finite) and the number of bandit arms (two or three)."

(Papadimitriou et al., 1994)

The complexity of optimal queueing network control

"We consider the classical problem of optimal control (routing and sequencing) of a network of queues. We prove that this problem is EXP-complete and, therefore, provably intractable. Similar results are established for restricted versions of the problem. A weaker result is also established for the restless bandit problem.<<ETX>>"

Rothschild experimentation and economic applications

Rothschild's contribution was to bring the bandit logic directly into economics: a firm can learn by taking costly actions today, such as setting prices, in order to improve future decisions. In this view, experimentation is not a side issue but part of optimal choice under incomplete information. (5 sources)

Rothschild's 1974 work is a standard economic use of the two-armed bandit idea. Later summaries describe it as using a bandit model to study market pricing and to ask how agents behave when demand is unknown. In the economic interpretation, a firm does not just passively face uncertainty; it actively learns by choosing prices and then observing demand, so current actions affect both profits now and information for later. (Branzei et al., 2019) (Charpentier et al., 2020) (Fryer et al., 2015)

This is the Rothschild experimentation problem in a simple form. A firm facing unknown demand can charge a price that seems best given current beliefs, or it can try a different price that may lower current profit but reveal more about the demand curve. Fryer et al. give this intuition very clearly: setting prices too high or too low is costly in the short run, but it can help the firm learn the optimal price for future periods. That is the same exploration-versus-exploitation tradeoff as in the basic two-armed bandit, but translated into an economic decision problem with incomplete information. (Fryer et al., 2015) (Charpentier et al., 2020)

This Rothschild-style use of bandits then spread well beyond pricing. Reviews note applications to clinical trials, product search, R&D, and firm strategy, all cases where a decision maker must balance immediate payoff against the value of information from trying uncertain actions. In management and organizational learning, bandit models inspired by Rothschild and Gittins are widely used to study adaptation in changing environments, again because they capture this same learning-by-doing problem in a compact way. (Fryer et al., 2015) (Li, 2021)

For the user's query, the key takeaway is that "Rothschild experimentation" usually means an economic version of risky learning under incomplete information: one repeatedly chooses among actions, the choice is itself informative, and information is costly because learning requires taking actions that may be worse in the short run. That framing is exactly what later work on strategic experimentation and dynamic monopoly learning builds on. (Charpentier et al., 2020) (Fryer et al., 2015) (Rady et al., 2009)

Evidence

(Branzei et al., 2019)

Multiplayer Bandit Learning, from Competition to Cooperation

> *"Rothschild (Rothschild, 1974) studied the theory of market pricing using a two-armed bandit model and asked the question of whether players eventually settle on the same arm."*

(Charpentier et al., 2020)

Reinforcement Learning in Economics and Finance

> *"A related problem is the one where an agent have to choose, repeatedly, among various options but with incomplete information. Multi-armed bandits come from one-armed bandit, understand slot machines, used in casinos. Imagine an agent playing with several one-armed bandit machines, each one having a different (unknown) probability of reward associated with. The game is seen as a sequence of single arm pull action and the goal is to maximize its cumulative reward. What could be the optimal strategy to get the highest return?"*

> *"The one-armed bandit problem was used in economics in (Rothschild, 1974), when trying to model the strategy of a single firm facing a market with unknown demand. In an extension, (Keller et al., 1999) consider the problem of the monopolistic firm facing an unknown demand that is subject to random changes over time."*

(Fryer et al., 2015)

Two-Armed Restless Bandits with Imperfect Information: Stochastic Control and Indexability

> *"The key idea in this class of models is that agents face a tradeoff between experimentation (gathering information on the returns to each arm) and exploitation (choosing the arm with the highest expected value)"*

> *".They have been used to analyze problems as diverse as market pricing, the optimal design of clinical trials, product search and the research and development activities of firms (Rothschild [66], Berry and Fristedt [7], Bolton and Harris [8], and Keller and Rady (Rady et al., 2009)). To understand how firms set prices without a clear understanding of their demand curves, Rothschild (Rothschild, 1974) posits that firms repeatedly charge prices and observe the resulting demand. Setting prices too high or too low is costly for firms (experimentation), but allows them to learn about the optimal price (exploitation)."*

(Li, 2021)

Strategic Exploration for Innovation

> *"identified the trade-off between exploration and exploitation in organization learning, multi-armed bandit models as in (Rothschild, 1974) and (Gittins, 1979) have been widely used in the management literature to study the exploration-exploitation trade-off in organizational adaption in dynamic environments; see, for example, (Denrell et al., 2001), (Posen et al., 2012), (Lee et al., 2016)Puranam (2016), or Stieglitz, Knudsen, and(Stieglitz et al., 2016)."*

(Rady et al., 2009)

Strategic Experimentation with Poisson Bandits

> *"In this paper we analyze a game of strategic experimentation where a finite number of players face identical two-armed bandit problems. There is a safe arm that offers a known and constant flow payoff and a risky arm whose lump-sum payoffs are driven by a Poisson process of unknown intensity. The risky arm can be either good or bad: if it is good, the lump sums arrive more frequently than if it is bad"*
>
> *".A single success on the risky arm does not fully reveal its type. This stands in marked contrast to the experimentation model of Keller et al. (2005)."*

Canonical two-armed risky learning under incomplete information

The canonical risky-learning model has one safe arm with a known payoff and one risky arm with an unknown type. The agent learns only by using the risky arm, so optimal behavior depends on prior beliefs, the signals generated by risky play, and how much future learning matters. (5 sources)

In the standard two-armed incomplete-information setup, one arm is safe and known, while the other is risky and informative. The safe arm gives a constant or known return, but the risky arm has an unknown payoff parameter or type, so choosing it creates both payoff risk and information about whether it is actually good or bad. This is the clean benchmark behind much of the later experimentation literature. (Cohen et al., 2009) (Marx et al., 2022) (Pourbabaee, 2020)

A common specification is binary uncertainty about the risky arm. The risky arm can be High or Low, good or bad, while the decision maker starts with a prior and updates that belief after observing outcomes from risky play. In this sense, “incomplete information” means not just uncertainty about payoffs, but uncertainty that is resolved endogenously through action. (Cohen et al., 2009) (Rady et al., 2009) (Pourbabaee, 2020)

The key asymmetry is that learning usually comes only from the risky arm. If the agent chooses the safe arm, she gets its known payoff but no new information about the uncertain alternative. Marx et al. state this point very directly: the untreated or safe arm is known at the outset, and pulling it yields no information about the treated or risky arm. That assumption makes experimentation sharply defined, because all belief updating is tied to taking the uncertain action. (Marx et al., 2022)

This setup appears in both discrete and continuous time, but the logic is the same. In continuous-time models, the decision maker may choose at each instant how much effort or resource to allocate to the safe versus risky arm, and the risky arm’s signals can come from Poisson arrivals or more general Lévy processes. These versions make the same underlying point: the agent trades off the sure current return from the safe arm against the option value of learning from the risky one. (Cohen et al., 2009) (Rady et al., 2009) (Klein et al., 2011)

A notable implication is that bad news can arrive slowly. In Poisson bandit models, even a success on the risky arm need not fully reveal that the arm is good, so beliefs often move gradually rather than collapsing after one observation. This makes experimentation a dynamic stopping problem: continue with the risky arm while beliefs remain high enough, and switch to the safe arm once posterior optimism falls below a cutoff. (Rady et al., 2009) (Klein et al., 2011) (Model-Generated)

For the user’s query, this is the canonical “two-armed bandit learning under incomplete information” model: one known safe action, one unknown risky action, Bayesian updating from risky play, and an optimal policy that values information because trying the risky arm can improve later choices even when it is not best in the short run. (Cohen et al., 2009) (Marx et al., 2022) (Pourbabaee, 2020)

Evidence

(Cohen et al., 2009)

Bandit problems with Levy payoff processes

"We study two-armed Levy bandits in continuous-time, which have one safe arm that yields a constant payoff s, and one risky arm that can be either of type High or Low; both types yield stochastic payoffs generated by a Levy process. The decision maker (DM) has to choose, at any given time t, the fraction of resource to be allocated to each arm over the time interval [t,t+dt)"

".In addition, we study two applications of the results: (a) we show how to price information in two-armed Levy bandit problem"

(Marx et al., 2022)

Parallel Trends and Dynamic Choices

"It is helpful to conceive of an individual's status in each period (treated/untreated) as "arms" (risky/safe) in a multi-armed bandit model. There is some learning about the mean returns from pulling these arms. That is, the decision-makers update beliefs based on past realized outcomes.

We start with a restriction that there is only learning from and about the treated arm.

In other words, the mean return of the untreated arm is known to decision-makers at the outset, and pulling the untreated arm yields no information about the treated arm. This corresponds to a simple two-armed bandit model with a safe and a risky arm."

(Pourbabaee, 2020)

Robust experimentation in the continuous time bandit problem

"At the core of our paper is an experimentation process between two projects framed as a two-armed bandit problem. The return rate to one arm is known to be r, whereas the return rate of the second arm is a binary random variable θ ∈ {θ, θ}, such that θ ≤ r ≤ θ.

The decision maker (henceforth DM) holds an initial prior p 0 = P θ = θ , that can be updated when she invests in the second project and learns its output. At the outset, she has to sequentially choose arms to learn about the unknown return rate while maximizing her net experimentation payoff."

(Rady et al., 2009)

Strategic Experimentation with Poisson Bandits

"In this paper we analyze a game of strategic experimentation where a finite number of players face identical two-armed bandit problems. There is a safe arm that offers a known and constant flow payoff and a risky arm whose lump-sum payoffs are driven by a Poisson process of unknown intensity. The risky arm can be either good or bad: if it is good, the lump sums arrive more frequently than if it is bad"

".A single success on the risky arm does not fully reveal its type. This stands in marked contrast to the experimentation model of Keller et al. (2005)."

(Klein et al., 2011)

Negatively Correlated Bandits

"We analyze a two-player game of strategic experimentation with two-armed bandits. Each player has to decide in continuous time whether to use a safe arm with a known payoff or a risky arm whose likelihood of delivering payoffs is initially unknown."

Costly information acquisition and value of experimentation

In these models, information is costly because the agent usually learns only by taking an action that may be worse right now. The value of experimentation is the future gain from that information, and it must be weighed against current payoff loss, explicit exploration costs, or forgone safe returns. (4 sources)

A useful way to state the issue is that information is not free: in the canonical two-armed problem, the agent often has to choose the risky arm in order to learn, and that choice can lower current payoff relative to the safe arm. So the value of experimentation is the expected future benefit from better decisions made after learning, net of the current cost of trying the uncertain option. This is exactly why experimentation can be optimal even when it looks unattractive in a one-period comparison.

Recent work makes this tradeoff very explicit by allowing direct exploration costs in addition to ordinary regret. In a two-armed setting, Yekkehkhany et al. model the case where “pulling an arm in the exploration phase has a cost” and propose a rule that balances a combined “cost-regret trade-off,” which they describe as corresponding to the exploration-exploitation tradeoff. That is a modern algorithmic version of the same economic idea: useful information may require paying for it now. (Yekkehkhany et al., 2021)

Experimental evidence also supports the idea that decision makers respond to this value of information. Hoelzemann et al. study a setting with one known-payoff arm and one unknown arm and find more experimentation when trying the unknown arm is initially optimal than when it is initially suboptimal. They interpret this as behavior that is broadly in line with normative predictions and conclude that subjects "value the information gained through costly experimentation." (Hoelzemann et al., 2018)

In economic applications, the cost of information acquisition often appears not as an explicit fee, but as an opportunity cost. In strategic experimentation models, a player who chooses the risky arm gives up the safe arm's known return while generating information that may help later choices. Das et al. make this point sharply by studying asymmetric safe-arm payoffs in a two-armed strategic bandit and interpreting these differences as differences in the "opportunity costs" of learning on the risky arm. They show that equilibrium cutoff behavior depends on how different these costs are across players, extending earlier results on asymmetry in strategic experimentation. (Das et al., 2024) (Das et al., 2019)

For the user's query, this is the key link between "risky learning," "costly information acquisition," and "Rothschild experimentation." Information has value because it improves future action, but getting that information usually requires taking the risky arm, paying an explicit cost, or giving up a safe payoff. The central problem is therefore not just whether to learn, but whether the expected gain from learning is large enough to justify that present sacrifice.

Evidence

(Yekkehkhany et al., 2021)

A cost–based analysis for risk–averse explore–then–commit finite–time bandits

> *"In the case that pulling an arm in the exploration phase has a cost, we propose the c-OTE-MAB algorithm for two-armed bandits that addresses the cost-regret trade-off, corresponding to exploration-exploitation trade-off, by minimizing a linear combination of cost and regret that is called cost-regret function, using a hyper-parameter."*

(Hoelzemann et al., 2018)

Bandits in the lab

> *"In their experimental design, the expected payoff of one arm is known, while the other is unknown. Experimentation is observed more in one treatment where initial selection of the unknown arm is optimal compared to the treatment where experimentation is suboptimal. These results suggest that subjects' behavior is consistent with the normative predictions and that subjects value the information gained through costly experimentation."*

(Das et al., 2024)

Strategic experimentation with asymmetric safe options

> *"There are many situations when information generated through exploration by one agent is helpful to other agents as well"*
>
> *".In terms of a two-armed bandit model, this means the safe-arm payoffs were different across players. In this article, we aim to analyse the impact of asymmetric safe-arm options in a game of strategic experimentation with two-armed exponential bandits"*
>
> *".In this article, we have considered a two-armed strategic bandit model where players face different opportunity costs of experimentation in terms of safe-arm payoffs but an identical rate of learning on the risky arm. We show that, as in Das, (Das et al., 2019), there exists a Markov Perfect equilibrium in cut-off strategies if and only if players are sufficiently asymmetric, i.e., the difference in safe-arm payoffs is high enough. Hence, we show that the conclusion that a sufficient degree of asymmetry across players is necessary and sufficient for the existence of an equilibrium in cutoff strategies continues to apply also when the asymmetry pertains to the opportunity costs, rather than the yield, of information acquisition via the risky arm."*

(Das et al., 2019)

Strategic experimentation with asymmetric players

> *"This paper contributes to the literature on strategic experimentation with bandits, a problem studied quite widely in economics, amongst others, by Bolton and Harris (1999), Keller et al. (2005), (Rady et al., 2009), (Klein et al., 2011), Klein (2013) and Thomas (2017)"*
>
> *".Many variants of this problem have been studied in the literature2 ."*

Learning dynamics, complete learning, and risk/horizon effects

Learning in two-armed bandits is path dependent: early samples matter a lot, and experimentation is more attractive when there is enough time left to use what is learned. Whether agents eventually learn the true arm, and how risk attitudes change experimentation, depends strongly on the horizon and on whether the model is discounted or undiscounted. (5 sources)

A useful way to think about learning dynamics is that the value of trying the risky or less-sampled arm is highest early on, when beliefs are still diffuse and there is still time to benefit from new information. In a finite-horizon two-armed task, the agent updates beliefs after each outcome, and an unsampled arm can have higher future expected value because it might turn out to be better, because the agent can switch back if it is disappointing, and because a long enough remaining horizon lets the agent recover the cost of sampling it. As Averbeck emphasizes, experimentation is therefore concentrated in the first few choices rather than spread evenly over time. (Averbeck, 2015)

This horizon effect is central for risky learning under incomplete information. If little time remains, information has low option value, so the safe arm becomes relatively more attractive; if much time remains, the payoff from learning can justify short-run losses from experimentation. In this sense, the stopping rule in canonical two-armed models is not just about current beliefs, but also about how many future decisions those beliefs can still improve. (Averbeck, 2015) (Model-Generated)

The horizon also matters for whether learning is eventually complete. A notable result summarized by Jacko is that complete learning arises in the undiscounted setting through the structure of the Gittins index rule, citing Kelly's earlier result. The broad implication is that if the decision maker does not discount future payoffs, there is stronger pressure to keep experimenting enough to resolve uncertainty, whereas discounting can lead the agent to stop early and live with residual uncertainty. (Jacko, 2019)

Risk attitudes add another layer. In standard intuition, risk aversion might seem to discourage use of the risky arm. But bandit models create a second effect: experimentation can reduce uncertainty over time, so taking the risky arm can also be a way to manage future risk. Keller et al., building on the Rothschild-style continuous-time tradition, stress this point directly and note that the reason for some counterintuitive comparative statics is that "risk in bandit models can be reduced through experimentation with the risky arm." (Keller et al., 2019) (Roberts et al., 1981)

That theoretical point is also consistent with experimental evidence that some subjects do gather information in ways that fit risk-sensitive experimentation rather than simple myopic avoidance of uncertainty. Fudenberg et al. report that many participants collect evidence and that some choices to avoid entry are consistent with risk aversion in an environment where experimentation is possible. This supports the idea that risk aversion does not simply shut down learning; instead, it can interact with the value of information in more subtle ways. (Fudenberg et al., 2019)

For the user's query, the main lesson is that risky learning in the Rothschild two-armed tradition is shaped by dynamics: agents experiment more when uncertainty is high and time is long, complete learning is more plausible when future payoffs are weighted heavily, and risk aversion can either dampen or sustain experimentation because trying the risky arm may itself be the way to reduce uncertainty for later decisions. (Averbeck, 2015) (Jacko, 2019) (Keller et al., 2019) (Fudenberg et al., 2019)

Evidence

(Averbeck, 2015)

Theory of Choice in Bandit, Information Sampling and Foraging Tasks

> *"We began by examining the explore-exploit trade-off in a stationary 2-armed bandit task, in which both bandits paid-off with the same fixed reward. The bandits varied, however, in the fraction of times they delivered a reward if chosen. In this case the explore-exploit trade-off affects the first few choices, before both targets have been sampled a few times. We modeled this as a finite state,*

finite horizon, undiscounted POMDP, where the information states were the number of times each bandit was chosen, C i and the number of times each bandit was rewarded, R i . This information state space is formed by the sufficient statistics for the two bandit processes. Transitions through the information state space occur after each choice and its associated outcome and they correspond to belief updates for the process"

".In the first few trials there is value to sampling both options, and unsampled options have a larger FEV. This future expected value depends on three factors. First, the distribution over possible reward probabilities for the unsampled option is broad, given by the prior. Thus, the unsampled option may be more rewarding than the options which have been sampled. Second, if the unsampled option is sampled, and it is not as good as the other options, the subject can switch back to the other options. However, if the (previously) unsampled option is better than the other options, the subject can stick with it. Finally, the time horizon must be long enough to reap the rewards of investing samples in the novel option."

(Jacko, 2019)

The Finite-Horizon Two-Armed Bandit Problem with Binary Responses: A Multidisciplinary Survey of the History, State of the Art, and Myths

"This is however true only in the discounted setting; (Kelly, 1981) proved that the structure of the Gittins index rule in the undiscounted setting (see subsubsection 5.3.2) leads to complete learning."

(Keller et al., 2019)

A note on optimal experimentation under risk aversion

"Following (Roberts et al., 1981) and (Bonatti et al., 2017), who in turn built upon the seminal work of Rothschild (1974a,b), it uses a continuous-time framework. We extend the standard model by allowing for risk aversion on behalf of the decision maker"

"The reason for this counterintuitive result relates to the notion that risk in bandit models can be reduced through experimentation with the risky arm."

(Roberts et al., 1981)

FUNDING CRITERIA FOR RESEARCH, DEVELOPMENT, AND EXPLORATION PROJECTS

"Department of Energy, nor any of their employees, nor any of their contractors, subcontractors, or their employees, makes any warranty, express or implied, or assumes any legal liability or responsibility for the accuracy, completeness, or usefulness of any information, apparatus, product or process disclosed or represents that its use would not infringe privately owned rights. ABSTRACT The sequential nature of activities like research, development, or exploration requires optimal funding criteria to take account of the fact that subsequent funding decisions will be made throughout the future. Thus, there is a continual possibility of reviewing a project's status, based on the latest information. After setting up a model to capture this feature, optimal funding criteria are investigated. In an important special case, an explicit formula is derived. As well as throwing light upon the nature of development activities, the analysis is also relevant to the general theory of information gathering processes. for energy research and development and more specifically for management decisions in the Photovoltaic Program."

(Fudenberg et al., 2019)

Learning Theory and Heterogeneous Play in a Signaling-Game Experiment

"To further examine the extent to which these choices are consistent with risk aversion in an environment with experimentation, we conducted additional treatments using a bandit problem. The main findings from these treatments, reported in the online Appendix, are that many participants collect evidence and respond in a manner consistent with the SCE, and that some participants' choices to stay Out when Blue in the random-type treatment are consistent with risk aversion."

Strategic, social, and correlated-information extensions

Once more than one agent is involved, experimentation is no longer just a private exploration problem. Each person's risky choice can create information for others, change incentives, and generate equilibria that differ from the single-agent cutoff rule. (6 sources)

A major extension of the Rothschild-style two-armed bandit is **strategic experimentation**, where several agents can learn from a common risky source or from the consequences of each other's actions. This literature is now broad in economics, with work building on continuous-time two-armed bandits and studying how equilibrium experimentation differs from the single-decision-maker benchmark. Das et al. describe strategic experimentation with bandits as a widely studied problem and point to many variants in the literature. (Das et al., 2019)

In these models, the main new issue is that one agent's experimentation can produce information that others

also benefit from, so private incentives to experiment need not match social incentives. A notable example is Rady et al.'s two-armed Poisson bandit game, where players share uncertainty about the risky arm and choose in continuous time between safe and risky actions. They show that equilibrium behavior need not take the simple all-cutoff form familiar from the single-agent case, and that equilibria exhibit an "encouragement effect" relative to the single-agent optimum. (Das et al., 2019) (Rady et al., 2009)

Another important extension changes the information structure itself by making risky payoffs **correlated across agents**. Klein et al. study a two-player two-armed bandit in which the risky arms are perfectly negatively correlated, and this sharply changes the learning result: unlike the common-type case, complete learning can arise in any Markov perfect equilibrium when stakes are high enough, and equilibrium behavior takes cutoff form. Depending on stakes, the model can yield a planner-like symmetric equilibrium, myopic behavior, or a continuum of equilibria. (Das et al., 2019) (Klein et al., 2011)

Correlation also appears in single-agent settings where trying one project reveals information about another. Pourbabaee makes this explicit by showing that a self-experimentation problem is formally equivalent to a two-armed bandit with correlated arm payoffs, with the correlation coming from the agent's hidden type. This matters for the user's query because it broadens "two-armed risky learning under incomplete information": the uncertain payoff of one arm may teach the agent about other choices too, not just about that one arm in isolation. (Pourbabaee, 2022)

A related branch studies **social learning from others' experimentation** rather than only one's own payoffs. Cheredina et al. place this next to the costly-information literature and note work where agents observe one another's risky choices but not the payoffs, so learning is indirect and stopping can be irreversible. Their summary also connects costly search to asymptotic learning results: when information is costly, responsiveness of actions to beliefs alone does not guarantee full long-run learning. (Cheredina et al., 2025)

Recent work also studies collective-choice versions of the two-armed bandit. Chen analyzes an N-agent model in which agents vote between a safe action and a risky action, and the risky arm is implemented only if support passes a threshold. In that setup, experimentation depends not just on beliefs and payoffs, but also on the decision rule that aggregates individual votes. This is another way strategic interaction can reshape the basic safe-versus-risky learning problem. (Chen, 2025)

For the user's query, these extensions show how the classic Rothschild experimentation problem changes once information is shared, inferred from others, or correlated across actions. The result is still a two-armed risky-learning problem under incomplete information, but now the value and cost of information acquisition depend on equilibrium behavior, information spillovers, and the structure of correlation, not just on one agent's private tradeoff.

Evidence

(Das et al., 2019)

Strategic experimentation with asymmetric players

> *"This paper contributes to the literature on strategic experimentation with bandits, a problem studied quite widely in economics, amongst others, by Bolton and Harris (1999), Keller et al. (2005), (Rady et al., 2009), (Klein et al., 2011), Klein (2013) and Thomas (2017)"*
>
> *".Many variants of this problem have been studied in the literature2 ."*

(Rady et al., 2009)

Strategic Experimentation with Poisson Bandits

> *"In this paper we analyze a game of strategic experimentation where a finite number of players face identical two-armed bandit problems. There is a safe arm that offers a known and constant flow payoff and a risky arm whose lump-sum payoffs are driven by a Poisson process of unknown intensity. The risky arm can be either good or bad: if it is good, the lump sums arrive more frequently than if it is bad"*

".A single success on the risky arm does not fully reveal its type. This stands in marked contrast to the experimentation model of Keller et al. (2005)."

(Klein et al., 2011)

Negatively Correlated Bandits

"We analyze a two-player game of strategic experimentation with two-armed bandits. Each player has to decide in continuous time whether to use a safe arm with a known payoff or a risky arm whose likelihood of delivering payoffs is initially unknown."

(Pourbabaee, 2022)

Reputation, learning and project choice in frictional economies

"The self-experimentation model that is developed in this paper is formally equivalent to a two-armed Bandit setting, where the arms' payoffs are correlated.Specifically, what correlates the payoffs of the projects is the single dimensional variable that represents the agent's hidden type."

(Cheredina et al., 2025)

Two Motives for Verification in Information Cascades

"Mueller-Frank and Pai (2016) let agents acquire information through costly search with heterogeneous private costs and obtain asymptotic learning ex-actly when costs are not bounded away from zero. (Kultti et al., 2006) shows more generally that responsiveness of actions to beliefs does not by itself restore asymptotic learning when information is costly"

". A related experimentation literature studies what agents learn from others' risky choices. Rosenberg et al. (2007) analyze two repeatedly acting players in a one-armed-bandit problem: players observe one another's actions but not payoffs, and stopping experimentation is irreversible."

(Chen, 2025)

Collective Experimentation with Correlated Payoffs

"Specifically, we analyze a two-armed exponential bandit model involving N agents. At each instant, the agents jointly decide between a safe action S and a risky action R under a predetermined collective decision rule: Each agent casts a vote for either R or S. If the share of votes for R exceeds a given threshold k (0, 1], then R is implemented; otherwise, S is implemented. The safe action S yields a constant, homogeneous payoff to each agent, while R yields intermittent payoffs depending on the agents' types, which are unknown initially."

Related but non-bandit or boundary cases

Not every dynamic learning problem with costly information is a bandit problem. A useful boundary case is when information gathered during experimentation helps choose a final action, but the payoffs from the experimental phase itself do not enter welfare. (1 source)

This boundary matters for the user's query because some papers look close to Rothschild experimentation but are not quite the same object. In the classic two-armed bandit, current rewards and future learning are jointly part of the objective, so using the risky arm trades off present payoff against information for later use. By contrast, some dynamic information-acquisition problems treat experimentation as only an input into a later decision, and do not count the payoffs from the experimental stage itself. Adusumilli makes this distinction very clearly, noting that when "in-sample outcomes are not included in the welfare calculation," the setup is different from a multi-armed bandit problem. (Adusumilli, 2022)

So, as a practical guide, a model is closest to the Rothschild two-armed bandit tradition when the agent repeatedly chooses between a safe and a risky option, learns from those choices, and directly cares about the stream of payoffs during learning. If instead the learning phase is only for data collection before a final one-shot choice, then it is better seen as a related sequential learning or experimental-design problem rather than a canonical bandit. This distinction helps keep the user's search terms sharp: "costly information acquisition" and "incomplete information" overlap with bandits, but not every such model is a two-armed risky-learning bandit in the Rothschild sense. (Adusumilli, 2022)

Evidence

(Adusumilli, 2022)

Neyman allocation is minimax optimal for best arm identification with two arms

> *"However, in-sample outcomes are not included in the welfare calculation, thus differentiating this setup from the multi-armed bandit problem."*